

39.5 Priors as regularizers

Proposition 30.1.4 already said that every penalty is a prior and every prior a penalty: the maximizer of a penalized log-likelihood is the mode of a posterior whose log prior is minus the penalty. This section takes that identity in the other direction and asks what a prior *does* in a generalized linear model. The answer is more than shrinkage: a prior carrying almost no information can still be the difference between an estimate and no estimate at all.

39.5.1 Two correspondences and a warning about scale

Proposition 39.5.1 — What a prior does

Let ℓ be the log-likelihood of a generalized linear model with $\text{rank}(\mathbf{X}) = p$ and a concave log-likelihood (Theorem 34.3.3), and write $\boldsymbol{\beta} = (\beta_0, \boldsymbol{\gamma}^\top)^\top$ with β_0 an intercept.

- (a) (*Normal prior, ridge penalty.*) Under $\gamma_j \sim N(0, \tau^2)$ independently and a flat prior on β_0 , the posterior mode maximizes $\ell(\boldsymbol{\beta}) - \frac{1}{2}\lambda\|\boldsymbol{\gamma}\|^2$ with $\lambda = 1/\tau^2$. If ℓ is bounded above, as it is for the binomial and the Poisson, and the response is not constant — for the binomial, both outcomes occur — the mode exists and is unique whether or not the data are separated. The hypothesis on the response cannot be dropped: with a flat prior on β_0 the objective is not coercive in β_0 when every y_i is 1. Penalizing the intercept as well makes existence unconditional.
- (b) (*Laplace prior, lasso penalty.*) Under γ_j independent with density $(2b)^{-1}e^{-|\gamma_j|/b}$ and a flat prior on β_0 , the posterior mode maximizes $\ell(\boldsymbol{\beta}) - \lambda \sum_j |\gamma_j|$ with $\lambda = 1/b$, and is therefore the lasso-penalized estimate of Definition 27.4.1, with exact zeros. The *posterior*, by contrast, is never sparse: whenever it is proper, $\Pr(\gamma_j = 0 \mid \mathbf{y}) = 0$ for every j .
- (c) (*What a normal prior says about a probability.*) In a binary model with logit link, suppose the linear predictor at some covariate value has the prior $\eta \sim N(0, \sigma^2)$. The induced prior density of $\pi = e^\eta/(1 + e^\eta)$ on $(0, 1)$ is unimodal with mode $1/2$ if $\sigma^2 \leq 2$, and has exactly two modes, symmetric about $1/2$, with a local minimum at $1/2$, if $\sigma^2 > 2$.
- (d) (*Existence.*) Under the prior of (a), or under independent Cauchy priors on the γ_j with a flat prior on β_0 , the logistic posterior mode exists for every data set in which both outcomes occur, separated or not; under (a) it is also unique. Under any proper prior the posterior itself is proper (Proposition 39.4.1(a)), so posterior means and credible intervals exist whether or not the maximum likelihood estimate does.

Proof.

- (a) The log posterior is $\ell(\boldsymbol{\beta}) - \|\boldsymbol{\gamma}\|^2/(2\tau^2)$ up to a constant, which is the stated penalized log-likelihood with $\lambda = 1/\tau^2$; this is Proposition 30.1.4 for the normal prior. Existence and uniqueness for the logistic model with the intercept left out of the penalty are Exercise (B1), under exactly the stated hypothesis; the same argument — a concave ℓ bounded above, coercive in β_0 at fixed $\boldsymbol{\gamma}$, plus a strictly concave

coercive quadratic — applies to any family whose log-likelihood is concave and bounded above. When $\boldsymbol{\gamma}$ is penalized and β_0 is not, and every y_i equals 1, the supremum 0 is approached only along $\boldsymbol{\gamma} = \mathbf{0}$, $\beta_0 \rightarrow +\infty$ and is not attained.

- (b) The log prior is $-\sum_j |\gamma_j|/b$ up to a constant, giving the penalty with $\lambda = 1/b$; this is Proposition 30.1.4 for the Laplace prior. For the second claim, the unnormalized posterior density is positive and continuous on $\mathbb{R}^p$, so when it integrates the marginal posterior of each γ_j is absolutely continuous and $\Pr(\gamma_j = 0 \mid \mathbf{y}) = 0$. The sparsity of the lasso is a property of the *mode* of a density with a kink, not of the distribution; Example (Posterior mode and posterior mean under a Laplace prior) makes the same point in the orthonormal linear model. Note what is *not* claimed: the posterior mean can be exactly zero, as it is whenever the posterior happens to be symmetric in γ_j about the origin.

- (c) Write $t = \log\{\pi/(1-\pi)\}$, so $dt/d\pi = 1/\{\pi(1-\pi)\} > 0$. The induced density satisfies

$$f(\pi) \propto \exp\left(-\frac{t^2}{2\sigma^2}\right) \frac{1}{\pi(1-\pi)},$$

with a constant of proportionality free of π . Since $t \mapsto \pi$ is strictly increasing, the stationary points of f in π are the stationary points of $\psi(t) = \log f(\pi(t))$ in t . Using $\pi(1-\pi) = e^t/(1+e^t)^2$,

$$\begin{aligned}\psi(t) &= -\frac{t^2}{2\sigma^2} + 2\log(1+e^t) - t + \text{const}, \\ \psi'(t) &= -\frac{t}{\sigma^2} + \frac{2e^t}{1+e^t} - 1 = \tanh(t/2) - \frac{t}{\sigma^2}.\end{aligned}$$

Call the last expression $G(t)$; it is odd and $G(0) = 0$. Now $G'(t) = \frac{1}{2} \text{sech}^2(t/2) - 1/\sigma^2$, which is even, strictly decreasing in $|t|$, and has maximum $\frac{1}{2} - 1/\sigma^2$ at $t = 0$. If $\sigma^2 \leq 2$ then $G' \leq 0$ everywhere, with equality only at $t = 0$ in the boundary case, so G is strictly decreasing and $t = 0$ is its only root: f has one stationary point. It is a maximum because $f(\pi) \rightarrow 0$ as $\pi \rightarrow 0$ or $\pi \rightarrow 1$, which follows from $\exp\{-t^2/(2\sigma^2) + |t|\} \rightarrow 0$.

If $\sigma^2 > 2$ then $G'(0) > 0$ and G' has exactly two zeros, so G has at most three roots; it has at least three, because $G(0) = 0$, $G > 0$ just to the right of zero while $G(t) \rightarrow -\infty$ as $t \rightarrow \infty$, and by oddness on the left. So there are roots $t_- < 0 = t_0 < t_+$ with $t_- = -t_+$. Since $f \rightarrow 0$ at both ends and f is continuous and positive, the outer two are maxima and the middle one, at $\pi = 1/2$, is a minimum.

- (d) With the normal prior the objective is coercive and strictly concave, as in (a). With independent Cauchy priors of scales s_j , the log posterior is $\ell(\boldsymbol{\beta}) - \sum_j \log\{1 + (\gamma_j/s_j)^2\}$ up to a constant. For the logistic model $\ell \leq 0$, and the penalty tends to $-\infty$ as $\|\boldsymbol{\gamma}\| \rightarrow \infty$; if $\boldsymbol{\gamma}$ stays bounded while $|\beta_0| \rightarrow \infty$ then $\ell \rightarrow -\infty$, provided both outcomes occur, by the argument of Exercise (B1). So the objective is coercive and continuous and attains its maximum on a compact sublevel set. It need not be unique: the Cauchy log density is not concave. The last sentence is Proposition 39.4.1(a).

■

Part (c) is the warning. A prior that is “noninformative” about a coefficient is a strong statement about a probability: a $N(0, 10^2)$ prior on the linear predictor puts half of its mass on probabilities

below 0.0012 or above $1 - 0.0012$, saying before any data that the event is nearly certain or nearly impossible and that the analyst does not know which. Panel (a) of [Figure 39.5.1](#) shows the three shapes: a gentle hump at $\sigma = 1$, flat at the boundary $\sigma = \sqrt{2}$, a deep U at $\sigma = 4$.

This is why the default priors recommended for logistic regression are much tighter than “vague”. Gelman, Jakulin, Pittau and Su (2008) rescale every covariate to mean zero and standard deviation 1/2, then put an independent Cauchy prior of scale 2.5 on each coefficient and 10 on the intercept. On that scale a coefficient of 5 multiplies the odds by $e^{2.5} \approx 12$ for a one standard deviation change, beyond anything social or biomedical data produce; the Cauchy’s heavy tail nonetheless leaves a genuinely large coefficient nearly alone, which a normal prior of the same scale would not.

39.5.2 Priors as penalties, checked

Remark (Scaling is not optional). Both correspondences put the same λ on every coefficient, which is a statement about the units of the covariates: multiplying x_j by ten divides γ_j by ten and changes its penalty by a factor of a hundred. Every penalized or weakly-informative fit therefore standardizes first (Proposition 27.2.1, Proposition 30.2.3). A prior of fixed scale is meaningful only once the covariates have one.

On the 944 respondents of Example (Voting intention in the 1996 American election), with the four covariates standardized, maximizing $\ell(\boldsymbol{\beta}) - \frac{1}{2}\lambda\|\boldsymbol{\gamma}\|^2$ by Newton’s method and maximizing $\ell(\boldsymbol{\beta}) - \lambda\sum_j |\gamma_j|$ by coordinate descent gives the two paths. At $\lambda = 64$, which is a normal prior with $\tau = 1/8$, the ridge estimates of the four slopes are 1.465, 0.105, 0.032 and 0.153; the lasso estimates are 1.773 and three exact zeros. Both have shrunk the party-identification coefficient from its unpenalized value of 2.769 on this standardized scale, the ridge further than the lasso, because the lasso has removed the competitors and the party term absorbs what they explained.

```
import numpy as np
import statsmodels.api as sm
from scipy import optimize, special, stats
anes = sm.datasets.anes96.load_pandas().data
Xf = np.column_stack([anes["PID"], anes["age"], anes["educ"], anes["income"]])
Xf = (Xf - Xf.mean(0)) / Xf.std(0)                                # standardize, as any
                                                                    # penalty requires
Xf = np.column_stack([np.ones(len(Xf)), Xf])
yf = anes["vote"].to_numpy(float)

def map_estimate(penalty, lam, steps=30):
    """Posterior mode: maximize the log-likelihood minus lam times a penalty on the
    slopes.

    A normal prior gives the ridge penalty and is solved by Newton's method; a
    Laplace
    prior gives the lasso penalty and is solved by coordinate descent on the working
    least squares problem of each Newton step.
    """
    beta = np.zeros(Xf.shape[1])
    for _ in range(steps):
        pi = special.expit(Xf @ beta)
        w = np.maximum(pi * (1 - pi), 1e-8)
        z = Xf @ beta + (yf - pi) / w                                # the working response
        if penalty == "ridge":
```

```

P = np.diag([0.0] + [1.0] * (Xf.shape[1] - 1))
beta = np.linalg.solve(Xf.T @ (w[:, None] * Xf) + lam * P, Xf.T @ (w * z
))
    continue
    for _ in range(40):                                # coordinate descent for the
        lasso
            for j in range(Xf.shape[1]):
                r = z - Xf @ beta + Xf[:, j] * beta[j]
                num, den = np.sum(w * Xf[:, j] * r), np.sum(w * Xf[:, j] ** 2)
                beta[j] = (num / den if j == 0
                           else np.sign(num) * max(abs(num) - lam, 0.0) / den)
    return beta

unpenalized = map_estimate("ridge", 0.0)                # the maximum likelihood fit
lams = np.array([1.0, 4.0, 16.0, 64.0, 256.0])
ridge_path = np.array([map_estimate("ridge", l) for l in lams])
lasso_path = np.array([map_estimate("lasso", l) for l in lams])
print(f"unpenalized slopes: {np.round(unpenalized[1:], 3)}")
print("normal prior with sd tau = 1/sqrt(lam) -> ridge; Laplace prior with b = 1/
    lam -> lasso")
for k, l in enumerate(lams):
    print(f" lam {l:6.1f} ridge {np.round(ridge_path[k, 1:], 3)}    "
          f"lasso {np.round(lasso_path[k, 1:], 3)}")

```

[Run this code online](#)

The Bayesian reading does not stop at the mode. Park and Casella (2008) noted that the Laplace prior is a scale mixture of normals — $\gamma_j \mid \tau_j^2 \sim N(0, \tau_j^2)$ with τ_j^2 exponential — which turns the lasso into a hierarchical model with normal full conditionals, and so into a Gibbs sampler whose output is the whole posterior rather than a sparse point. Priors designed to be *more* selective make the mixing distribution heavy-tailed at both ends: a spike-and-slab prior mixes a point mass at zero with a diffuse component, and the horseshoe of Carvalho, Polson and Scott (2010) takes τ_j half-Cauchy, so small coefficients are shrunk almost to zero and large ones almost not at all. The other penalties of Chapter 30 carry over by the same substitution, penalized least squares becoming penalized log-likelihood: the elastic net of Definition 30.2.1, the group lasso of Definition 30.3.1 for a factor's dummy variables, and the componentwise boosting of Definition 30.4.1 with the negative log-likelihood as loss and one Newton step per round.

39.5.3 A prior that rescues an estimate

Example. (The separated subgroup, three priors)

Example (A separated subgroup) fitted the intended vote on party identification among the 13 respondents whose schooling stopped at or before the eighth grade, and found complete separation: no maximum likelihood estimate exists, and software reports whatever coefficient its iteration limit produced. With only two parameters the posterior can be computed by quadrature, under any prior at all — over the whole plane, by the substitution $\beta = s \tan u$, since under separation a Cauchy posterior has a polynomial tail and any fixed box would discard a large part of it.

Under the flat prior the posterior is not a distribution at all, and the arithmetic shows it: integrating the likelihood over a square box centred at the origin gives 8.7 for half-width

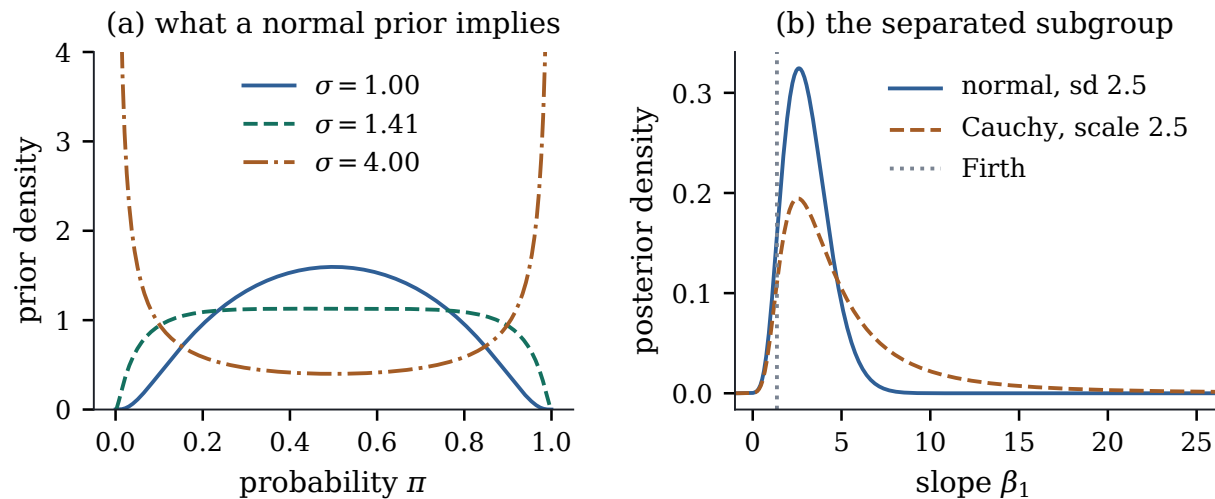


Figure 39.5.1: (a) The prior induced on a probability by a $N(0, \sigma^2)$ prior on the logit; the middle curve is the boundary case $\sigma^2 = 2$ of Proposition 39.5.1(c). (b) Posterior densities of the party-identification slope in the separated subgroup of Example (A separated subgroup) under a normal and a Cauchy prior, with Firth's estimate marked.

ten and 778.6 for half-width eighty, each doubling multiplying the mass by at least 4.13. That is the area of the separating cone growing, as in the proof of Proposition 39.4.1(b), and no amount of sampling will reveal it — a chain started near the data's own scale looks well behaved while wandering up a ridge of infinite mass.

An independent $N(0, 2.5^2)$ prior on the slope and $N(0, 10^2)$ on the intercept gives a proper posterior with mean 3.077, median 2.931 and a central 95 per cent credible interval of (1.046, 5.922). Independent Cauchy priors of the same scales give mean 6.091, median 4.001 and interval (1.127, 22.995). Both exclude zero decisively. Both lower endpoints agree to within a tenth, and the upper endpoints differ by a factor of 3.9: the data bound the effect from below, and everything above that is prior. Panel (b) of Figure 39.5.1 shows the two posteriors.

Set these beside the frequentist answers of Section 35.3. The ordinary likelihood ratio gives the one-sided interval $(0.902, \infty)$: the most honest statement available without a prior, and the least useful. Firth's penalty, the mode under the Jeffreys prior (Proposition 35.3.2), gives 1.362 with the penalized likelihood ratio interval (0.364, 4.045) — smaller than either posterior median, and a mode rather than a mean. The Jeffreys penalty falls off exponentially along the separating ray (Proposition 35.3.2) while a Cauchy of scale 2.5 falls off only polynomially, so it bites hardest on exactly the direction the data do not constrain. There is no way to choose among these numbers from the data. Thirteen observations that separate perfectly contain a bound and nothing more, and every method reports that bound plus whatever it assumed. The discipline is to say which assumption produced which number, and to notice that the flat prior is not the assumption-free option but the one whose assumption is inconsistent.

```
sub = anes[anes["educ"] == 1]                                # schooling stopped at or
    before grade eight
Xs = np.column_stack([np.ones(len(sub)), sub["PID"]])
ys = sub["vote"].to_numpy(float)

def loglik(b0, b1):
```

```

"""Logistic log-likelihood at one intercept and a whole grid of slopes."""
eta = b0 + np.outer(b1, Xs[:, 1])
return np.sum(ys * eta - np.logaddexp(0.0, eta), axis=1)

def log_prior(b0, b1, kind, scale=2.5):
    if kind == "flat":
        return 0.0 * b0
    if kind == "normal":
        return -(b0 / 10.0) ** 2 / 2 - (b1 / scale) ** 2 / 2
    return -np.log1p((b0 / 10.0) ** 2) - np.log1p((b1 / scale) ** 2) # Cauchy

def posterior_of_slope(kind, k=1200, s0=30.0, s1=6.0):
    """Marginal posterior of the slope, by quadrature over the whole plane."""
    t, wt = np.polynomial.legendre.leggauss(k)
    u, wu = t * np.pi / 2, wt * np.pi / 2
    b0, b1 = s0 * np.tan(u), s1 * np.tan(u)
    j0, j1 = s0 / np.cos(u) ** 2, s1 / np.cos(u) ** 2
    logq = np.array([loglik(a, b1) for a in b0]) + log_prior(b0[:, None], b1[None, :], kind)
    q = np.exp(logq - logq.max())
    marg = ((j0 * wu)[:, None] * q).sum(axis=0) # integrate out the intercept
    dens = marg / np.sum(marg * j1 * wu) # density with respect to b1
    return u, b1, dens, j1, wu

def summarize(u, b1, dens, j1, wu, probs=(0.025, 0.5, 0.975)):
    """Posterior mean and quantiles of the slope from the quadrature grid."""
    f = dens * j1 # the density in the u coordinate
    cdf = np.concatenate([[0.0], np.cumsum((f[1:] + f[:-1]) / 2 * np.diff(u))])
    return np.sum(b1 * f * wu), np.interp(probs, cdf / cdf[-1], b1)

post_n, post_c = posterior_of_slope("normal"), posterior_of_slope("cauchy")
mean_n, (lo_n, med_n, hi_n) = summarize(*post_n)
mean_c, (lo_c, med_c, hi_c) = summarize(*post_c)
print(f"posterior mean of the slope: normal {mean_n:.3f}, Cauchy {mean_c:.3f}")

```

[Run this code online](#)

Idea. A penalty is a prior and a prior a penalty, but the two readings answer different questions. The penalty reading asks what estimate to report and tunes λ by cross-validation; the prior reading asks what is believed before the data, and its scale must be defensible on the scale of the parameter — for a binary response, as a statement about probabilities.

39.5.4 Exercises

39.5.4.1 A. Check your understanding

Exercise A1.

A colleague reports a ridge-penalized logistic fit with $\lambda = 100$ on standardized covariates. What normal prior does that correspond to, and what odds ratio for a one standard deviation change does the prior regard as, say, two standard deviations from zero?

Solution. $\lambda = 1/\tau^2$ gives $\tau = 0.1$. Two prior standard deviations is $\gamma_j = 0.2$, an odds ratio of $e^{0.2} = 1.22$. The prior therefore says that a one standard deviation change in any covariate almost certainly multiplies the odds by less than about 1.22: a strong belief, not a weak one, which may be appropriate in a high-dimensional screening problem and absurd in a four-covariate model.

Exercise A2.

Explain in words why the induced prior of [Proposition 39.5.1\(c\)](#) becomes U-shaped for large σ , without using the calculus of the proof.

39.5.4.2 B. Practice**Exercise B1.**

Show that a Cauchy prior with scale s is a scale mixture of normals: $\gamma \mid \lambda \sim N(0, \lambda)$ with λ having an inverse gamma distribution with shape $1/2$ and rate $s^2/2$. Explain how this turns a Cauchy prior into a Gibbs sampler with normal full conditionals in the probit model.

Solution. If $\lambda \sim \text{IG}(1/2, s^2/2)$ then the marginal density of γ is

$$\begin{aligned} \int_0^\infty \frac{1}{\sqrt{2\pi\lambda}} e^{-\gamma^2/(2\lambda)} \frac{(s^2/2)^{1/2}}{\Gamma(1/2)} \lambda^{-3/2} e^{-s^2/(2\lambda)} d\lambda \\ \propto \int_0^\infty \lambda^{-2} e^{-(\gamma^2+s^2)/(2\lambda)} d\lambda \propto \frac{1}{\gamma^2 + s^2}, \end{aligned}$$

the Cauchy density of scale s , the last step by the substitution $v = 1/\lambda$. In the Albert–Chib scheme of [Proposition 39.4.5](#), conditioning on the λ_j as well as on $\mathbf{z}$ makes the prior on $\boldsymbol{\gamma}$ normal with covariance $\text{diag}(\lambda_j)$, so the full conditional of $\boldsymbol{\beta}$ is normal, and that of each λ_j is inverse gamma with shape 1 and rate $(\gamma_j^2 + s^2)/2$: every step is a standard draw.

Exercise B2.

For $\sigma = 10$, compute the prior probability that $\pi = e^\eta/(1+e^\eta) < 0.01$ when $\eta \sim N(0, \sigma^2)$, and compare with $\sigma = 1$. Relate the answers to [Proposition 39.5.1\(c\)](#).

Exercise B3.

[Proposition 35.3.2](#) adds $\frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{W} \mathbf{X})$ to the log-likelihood. Show that this is the log of the Jeffreys prior density for the logistic model, so Firth's estimate is a posterior

mode. Why does that mode exist even under separation, when the maximum likelihood estimate does not?

Solution. The Jeffreys prior is $\propto \det\{\mathcal{I}(\boldsymbol{\beta})\}^{1/2}$ for the expected information $\mathcal{I}$, which for the logistic model is $\mathbf{X}^\top \mathbf{W} \mathbf{X}$ with $\mathbf{W} = \text{diag}\{\pi_i(1 - \pi_i)\}$ at $\boldsymbol{\beta}$ (Proposition 35.1.3); its logarithm is $\frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{W} \mathbf{X})$ up to a constant, so adding it to ℓ gives Firth's penalized log-likelihood and its maximizer is the posterior mode under that prior. It exists under separation for the reason Proposition 35.3.2 proves: along a separating ray $t\mathbf{b}$ the weights decay like $e^{-t\delta}$, so the log prior falls linearly in t while the log-likelihood, already at most zero, has nothing left to gain. The likelihood alone is flat along that ray; the prior supplies the curvature.

39.5.4.3 C. Going deeper

Exercise C1.

Redo Proposition 39.5.1(c) for the probit link: with $\eta \sim N(0, \sigma^2)$ and $\pi = \Phi(\eta)$, show that the induced density of π is unimodal when $\sigma^2 \leq 1$ and bimodal when $\sigma^2 > 1$, and say why the threshold differs from the logit's.